

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Display
Manufacturer: Samsung
Firmware Version: N/A
Model(s): QM32C, QM43C, QM50C, QM55C, QM65C, QM75C, QM85C, QB65C-N, QB75C-N, QB85C-N, QH43C, QH50C, QH55C, QH65C, QH75C, QB43C, QB50C, QB55C, QB65C, QB75C, QB85C, QB55C-N, QM98C, QH98C

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.18.0001-b002	2.23.0.13

Version History

Module Version	Date	Notes
1_0_4_0	12/11/2024	Added Notes for Device. Added models: QM32C, QM98C, QH98C.
1_0_1_0	6/5/2024	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- DeviceID variable must be set accordingly. Default value is '0'. DeviceID ranges from '0' to '224', and Broadcast. If Device ID is set to 'Broadcast', status is unavailable.
Example: `InterfaceName.DeviceID = '0'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='QM43C')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='QM43C')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 1515, Model='QM43C')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value '16:9' 'Original Ratio'	Value '4:3' '21:9'	Value 'Custom'
# AspectRatio example InterfaceName.Set('AspectRatio', '16:9')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Input	Value 'MagicInfo' ¹ 'DisplayPort'	Value 'HDMI 1' 'HDMI 3'	Value 'HDMI 2'
# Input example InterfaceName.Set('Input', 'MagicInfo')			
Command MenuNavigation	Value 'Menu' 'Up' 'Left'	Value 'Exit' 'Down' 'Enter'	Value 'Return' 'Right'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

¹ Only available for models that contain the MagicInfo function.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value '16:9' 'Original Ratio'	Value '4:3' '21:9'	Value 'Custom'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command Input	Value 'MagicInfo' ¹ 'HDMI 2' 'HDMI 3'	Value 'HDMI 1' 'HDMI 2 PC' 'HDMI 3 PC'	Value 'HDMI 1 PC' 'DisplayPort'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

Command	Value	Value
VideoMute	'On'	'Off'
# VideoMute example InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)		
Command	Value	
Volume	0 to 100 in steps of 1	
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)		

¹ Only available for models that contain the MagicInfo function.

Cable and Adapter Requirements

Captive Screw to 3.5mm TRS RS-232 Serial Cable

Notes for the Device

To ensure the functionality of the module, please set the following:

- Menu → System → Power Control → Remote Configuration → Active
- Menu → System → Security → Remote Management → Allow
- Home → Settings → ID Settings → PC Connection Cable → RS232C Cable

Serial communication

Port Type: RS-232

Baud Rate: 9600

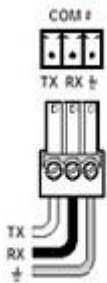
Data Bits: 8

Parity: None

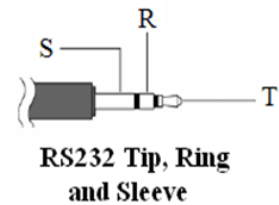
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
RXD	←	T	TXD
TXD	→	R	RXD
GND	→	S	GND



Network Communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 1515

Logon Credentials Supported: No

Multi-Connection Capabilities: Undetermined

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

To ensure the functionality of the module, please set the following:

- Menu → System → Power Control → Remote Configuration → Active
- Menu → System → Security → Remote Management → Allow
- Menu → System → Security → Secured Protocol → Off
- Menu → System → Security → Network Lock → Off
- Home → Settings → ID Settings → PC Connection Cable → RJ-45 (LAN)/Wi-Fi Network or RJ-45 (LAN) Cable

Appendix A. Set Commands (Device ID: 0)

Aspect Ratio 16:9	\xAA\x15\x00\x01\x10&
Aspect Ratio 21:9	\xAA\x15\x00\x01!7
Aspect Ratio 4:3	\xAA\x15\x00\x01\x18.
Aspect Ratio Custom	\xAA\x15\x00\x01"8
Aspect Ratio Original Ratio	\xAA\x15\x00\x01 6
Audio Mute Off	\xAA\x13\x00\x01\x00\x14
Audio Mute On	\xAA\x13\x00\x01\x01\x15
Executive Mode Off	\xAA]\x00\x01\x00^
Executive Mode On	\xAA]\x00\x01\x01_
Input DisplayPort	\xAA\x14\x00\x01%:
Input HDMI 1	\xAA\x14\x00\x01!6
Input HDMI 2	\xAA\x14\x00\x01#8
Input HDMI 3	\xAA\x14\x00\x011F
Input MagicInfo	\xAA\x14\x00\x01 5
Menu Navigation Down	\xAA\xB0\x00\x01a\x12
Menu Navigation Enter	\xAA\xB0\x00\x01h\x19
Menu Navigation Exit	\xAA\xB0\x00\x01- \xDE
Menu Navigation Left	\xAA\xB0\x00\x01e\x16
Menu Navigation Menu	\xAA\xB0\x00\x01\x1A\xCB
Menu Navigation Return	\xAA\xB0\x00\x01X\x09
Menu Navigation Right	\xAA\xB0\x00\x01b\x13
Menu Navigation Up	\xAA\xB0\x00\x01` \x11
Power Off	\xAA\x11\x00\x01\x00\x12
Power On	\xAA\x11\x00\x01\x01\x13
Video Mute Off	\xAA\xF9\x00\x01\x00\xFA
Video Mute On	\xAA\xF9\x00\x01\x01\xFB
Volume 0	\xAA\x12\x00\x01\x00\x13
Volume 100	\xAA\x12\x00\x01dw

Appendix B. Update Commands (Device ID: 0)

Aspect Ratio	\xAA\x15\x00\x00\x15
Audio Mute	\xAA\x13\x00\x00\x13
Executive Mode	\xAA]\x00\x00]
Input	\xAA\x14\x00\x00\x14
Power	\xAA\x11\x00\x00\x11
Video Mute	\xAA\xF9\x00\x00\xF9
Volume	\xAA\x12\x00\x00\x12